Complete Crouzeix Inequality: Exact Faber Metrics, Defect Accounting, and Certified Coefficient Cutoffs

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Abstract

This report examines the actual exterior-Faber and Hilbert–Schmidt amplification proof in the September 26, 2026 OpenAI manuscript, not a scalar substitute for its complete inequality. We reconstruct its positive two-marginal kernel, ordered polynomial tests, reciprocal coefficient weights, positive-block comparison, and treatment of degenerate numerical ranges. The local contribution is an exact nonnegative defect identity retaining cross energy, block slack, and the inverse of the full Faber boundary Gram operator. It gives conditional strict bounds and necessary near-extremality estimates. A finite-section theorem retains both resolvent and Faber-kernel tails and converts certified coefficient data into a valid upper bound; an exact Schur identity explains the missing complement in a naive finite inverse. Disk and ellipse calculations expose sharpness and nontrivial infinite tails. Selected private original proofs are used only for their physical-metric and retained/complement methodology, with their actual read ranges recorded. No universal improvement below two, independent global certification, novelty claim, or numerical certification by floating-point tests is asserted. Contribution assessment: C3/5, source-specific proved defect and tail certificates extend the audited proof quantitatively without claiming a new universal constant.

1 Scope, Status, and Exact Problem

For positive integers n, m, a matrix $A \in M_n(\mathbb{C})$, and $F(z) = \sum_{j=0}^{d} z^j C_j$ with $C_j \in M_m(\mathbb{C})$, put

$$W(A) = \{x^*Ax : \|x\| = 1\}, \qquad F[A] = \sum_{j=0}^{d} A^j \otimes C_j. \tag{1}$$

The base-matrix factor is first. The source claims, uniformly in both finite dimensions,

$$\|F[A]\| \le 2 \max_{z \in W(A)} \|F(z)\|. \tag{2}$$

All unmarked matrix norms are Euclidean operator norms; the Hilbert–Schmidt norm is *unnormalized*, $\|B\|_{\mathrm{HS}}^2 = \mathrm{tr}(B^*B)$. Circle measure is $d\tau(e^{it}) = dt/(2\pi)$. Changing either normalization midway changes the proof. The coefficient pairing below is bilinear, whereas Hilbert-space inner products are conjugate-linear in their first argument.

The source is OpenAI's *A direct proof of the complete Crouzeix inequality* [1]. Its entire main file, references, and figure source were read. The inherited argument is reconstructed in Sections 2–4; the new local results are Sections 5–6. This is a bounded mathematical review with no verified error found in the deductions checked, not a blanket certification of the release or all cited literature. In particular, the companion structural manuscript is not a dependency of the proof given here and was not independently audited for this report.

1.1 Public status, checked October 8, 2026

Lorist and Schwenninger's scalar result is arXiv:2608.03841v2, revised August 17, 2026 [2]. Their dilation argument requires commutativity and does not itself supply arbitrary matrix amplification. The separate paper by Åhag, Czyż, and Virtanen is arXiv:2608.27346v4, revised September 21, 2026 [3]; its Theorem 1.1 concerns base order at most three, with arbitrary coefficient size. Its higher-dimensional discussion is not an endorsement of the later OpenAI claim. The established earlier complete bound $1+\sqrt{2}$ is due to Crouzeix and Palencia [4]. This report does not conflate those three scopes.

The local frozen upstream audit reports that all five M630 blobs are unchanged between commits `adc7f1241b42e322a6451854ab7e4b4c146bf78a` and `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. These comprise four text files (including the figure source) and `paper.pdf`; only the text files were locally read. The latter commit was its October 8 public-head observation. A fresh public README request also returned the September 26 title and date. These are provenance observations, not proofs of mathematical correctness or permanent statements about a moving public branch.

2 Exterior Coordinates and the Exact Faber Metric

Throughout Sections 2–3, let Ω be a bounded convex domain with regular real-analytic Jordan boundary. Assume an exterior map h is meromorphic and one-to-one, including infinity, on $|\lambda|>r$ for some $r<1$, maps $|\lambda|>1$ onto the exterior of $\overline{\Omega}$, and has expansion

$$h(\lambda)=c\lambda+c_0+O(\lambda^{-1}),\quad c\neq 0,\qquad q(\lambda)=\lambda h'(\lambda). \tag{3}$$

Section 4 explains existence under the stated boundary hypotheses. On $\mathbb{T}$, q points outward and

$$\operatorname{Re}(\overline{q(\lambda)}(h(\lambda)-z))\geq 0\quad (z\in\overline{\Omega}). \tag{4}$$

Lemma 2.1 (Inherited mixed kernel construction)**.** *Define b_k at infinity by*

$$\frac{q(\lambda)}{h(\lambda)-z}=\sum_{k\geq 0} b_k(z)\lambda^{-k}. \tag{5}$$

Then $b_0=1$, b_k has degree k and leading coefficient c^{-k}, and

$$s(\lambda,\mu)=\frac{q(\lambda)}{h(\lambda)-h(\mu)}-\frac{\lambda}{\lambda-\mu}=\sum_{k,j\geq 1} s_{kj}\lambda^{-k}\mu^{-j}, \tag{6}$$

$$b_k(h(\mu))=\mu^k+\sum_{j\geq 1} s_{kj}\mu^{-j}\quad (k\geq 1). \tag{7}$$

The mixed series converges normally on the product collar, including both infinities and the diagonal. For each $r<\rho<1$ there is $C_\rho<\infty$ with $|s_{kj}|\leq C_\rho\rho^{k+j}$. On $\mathbb{T}^2$,

$$K=1+s+\overline{s}\geq 0,\qquad \int K(\lambda,\mu)\,d\tau(\lambda)=\int K(\lambda,\mu)\,d\tau(\mu)=1. \tag{8}$$

Proof. Expansion after factoring $c\lambda$ gives the degree and leading coefficient. For the joint assertion put $u = 1/\lambda$, $v = 1/\mu$, and write $h(\lambda) = c\lambda + g(u)$. The divided difference

$$H(\lambda, \mu) = \frac{h(\lambda) - h(\mu)}{\lambda - \mu} = c - uv\frac{g(u) - g(v)}{u - v}$$

is holomorphic on the reciprocal bidisk. It is nonzero off the diagonal by injectivity, on the diagonal by $h' \neq 0$, and on either axis because its value there is c. Thus $s = \lambda\partial_\lambda H/H = -u\partial_u H/H$ is holomorphic and vanishes on both reciprocal axes. Its Taylor expansion has only strictly mixed powers. Cauchy's estimate on the closed bidisk of radius ρ^{-1} gives the stated bound, which also proves normal convergence by a geometric majorant on each smaller compact set. Comparing coefficients in λ^{-1} gives (7); no boundary singularity is differentiated. For distinct circle points, $2\,\mathrm{Re}(\lambda/(\lambda - \mu)) = 1$. Hence $K = 2\,\mathrm{Re}(q(\lambda)/(h(\lambda) - h(\mu))) \geq 0$ by (4). Continuity handles the diagonal. Every term of s and $\bar{s}$ has nonzero frequency in *each* variable, proving the two separate marginals. □

Let $\mathcal{H}_m = \ell^2(\{0, 1, \ldots\}; M_m(\mathbb{C})_{\mathrm{HS}})$. On the positive-index part define

$$(SC)_j = \sum_{k\geq 1} s_{kj} C_k, \qquad S_{jk} = s_{kj}. \tag{9}$$

The reversal of the displayed coefficient indices is intentional. S acts on the index coordinate and identically on all matrix entries. Set

$$\mathcal{H} = 1 \oplus (I + S^*S), \qquad \mathcal{D} = 1 \oplus 2I. \tag{10}$$

The same symbols denote the scalar-index operators and their tensor extensions.

Proposition 2.2 (Exact metric and reciprocal norm)**.** *One has $\|S\| \leq 1$ and $I \preceq \mathcal{H} \preceq \mathcal{D} \preceq 2I$. For every finite sequence C and $G = \sum b_k C_k$,*

$$\int_{\mathbb{T}} \|G(h(\mu))\|_{\mathrm{HS}}^2 \, d\tau = \|C_0\|_{\mathrm{HS}}^2 + \sum_{k\geq 1} \|C_k\|_{\mathrm{HS}}^2 + \|SC_+\|^2 = \langle C, \mathcal{H}C\rangle. \tag{11}$$

If $Z \in \mathcal{H}_m$, the squared norm of the bilinear functional $\ell_Z(C) = \sum_{k,i,j} (Z_k)_{ij}(C_k)_{ij}$ in this metric is

$$Q_H(Z) = \langle \overline{Z}, \mathcal{H}^{-1}\overline{Z}\rangle. \tag{12}$$

Here and below the overbar is entrywise, not the matrix adjoint.

Proof. The integral operator $Tu(\mu) = \int K(\lambda, \mu)u(\lambda)\, d\tau(\lambda)$ is a contraction on Hilbert-valued L^2: Jensen uses the first marginal, then integration uses the second. For $u(\lambda) = \sum_{k\geq 1} C_k\lambda^k$, the strictly mixed Fourier series gives $Tu(\mu) = \sum_{j\geq 1}(SC)_j\mu^{-j}$. Parseval proves $\|S\| \leq 1$ first on finite sequences and then by density. Equation (7) separates constant, positive, and negative Fourier modes, proving (11). Finally $\ell_Z(C) = \langle \bar{Z}, C\rangle$ and the Riesz vector in the $\mathcal{H}$ metric is $\mathcal{H}^{-1}\bar{Z}$. Cauchy–Schwarz gives (12) with equality on that vector. Finite sequences suffice for the same supremum by density and equivalence of the two norms. This justifies the infinite dual without assuming that its maximizing vector is a polynomial. □

The source uses only the two inequalities obtained by deleting the last term in (11) or bounding it by $\|C_+\|^2$. We will retain the exact metric, but not attribute the standard Riesz or Schur algebra itself as a new theorem of functional analysis.

3 The Complete Amplification Argument

Assume now $W(A) \subset \Omega$; compactness makes this strict containment. Define

$$R(\lambda) = q(\lambda)(h(\lambda)I - A)^{-1}. \tag{13}$$

The eigenvalues of A belong to $W(A)$. The resolvent is therefore analytic on an exterior collar of $\mathbb{T}$ and at infinity, with expansion

$$R(\lambda) = \sum_{k\geq 0} b_k(A)\lambda^{-k}, \quad b_0(A) = I, \quad \|b_k(A)\| \leq C_R\theta^k \quad (0 < \theta < 1). \tag{14}$$

To obtain the final bound, select a circle $|\lambda| = \theta$ inside a collar free of resolvent poles and use Cauchy's coefficient formula; C_R can be the maximum of $\|R\|$ there. The collar and constants depend on A, h. For $B = h(\lambda)I - A$,

$$B^*(R + R^*)B = qB^* + \bar{q}B \succeq 0$$

by (4) applied to x^*Ax. Congruence yields $R + R^* \succeq 0$; its integral is $2I$. This proves the resolvent positivity and convergence needed for every subsequent sum and integral.

Theorem 3.1 (Inherited analytic-domain inequality)**.** *If* $\max_{\overline{\Omega}}\|F\| \leq 1$*, then* $\|F[A]\| \leq 2$*, uniformly in* n, m*. No scalar-to-complete inference is used.*

Proof. Write $\gamma = \|F[A]\|$. The case $\gamma = 0$ is immediate. Choose unit singular vectors with $F[A]x = \gamma y$ and $F[A]^*y = \gamma x$. Express $x = \sum_i Xe_i \otimes e_i$ and $y = \sum_i Ye_i \otimes e_i$, so X, Y are n by m matrices with $\|X\|_{\mathrm{HS}} = \|Y\|_{\mathrm{HS}} = 1$. Set

$$P = X^*RX, \quad Q = Y^*RY, \quad M = Y^*RX, \quad L = X^*RY, \qquad a^2 = \sum_{k\geq 0}\|M_k\|_{\mathrm{HS}}^2. \tag{15}$$

Subscripts denote Laurent coefficients. All four sequences are square summable by (14); $P_0 = X^*X$, $Q_0 = Y^*Y$ are Hermitian, and $L_0^* = M_0$.

For every matrix polynomial $G = \sum b_kC_k$, expansion in the fixed tensor order gives

$$y^*G[A]x = \sum_{k,i,j}(M_k)_{ij}(C_k)_{ij}, \qquad |y^*G[A]x| \leq a\Big(\sum_k\|C_k\|_{\mathrm{HS}}^2\Big)^{1/2}. \tag{16}$$

Indeed the ij term is $(Ye_i)^*b_k(A)Xe_j(C_k)_{ij}$. Taking $G = F$ proves $a > 0$. This calculation would be wrong if an unannounced adjoint or transpose were inserted in the coefficient pairing.

Evaluation preserves each ordered product, since $(A^i \otimes D_i)(A^j \otimes E_j) = A^{i+j} \otimes D_iE_j$. Consequently

$$y^*(FG)[A]x = \gamma x^*G[A]x, \qquad y^*(GF)[A]x = \gamma y^*G[A]y. \tag{17}$$

Use (16) on the *full* finite Faber expansions of FG and GF. The lower coefficient inequality from (11) and $\|FG\|_{\mathrm{HS}} \leq \|F\|\|G\|_{\mathrm{HS}}$, $\|GF\|_{\mathrm{HS}} \leq \|G\|_{\mathrm{HS}}\|F\|$ give

$$\max\{|x^*G[A]x|, |y^*G[A]y|\} \leq \frac{a}{\gamma}\|G \circ h\|_{L^2(\mathrm{HS})}. \tag{18}$$

The respective bilinear coefficient arrays are P_k and Q_k. Replacing $\mathcal{H}$ by $\mathcal{D}$ and testing finitely with $C_0 = \bar{P}_0$, $C_k = \bar{P}_k/2$ for $1 \leq k \leq N$, proves

$$D_P := \|P_0\|_{\mathrm{HS}}^2 + \tfrac{1}{2}\sum_{k\geq 1}\|P_k\|_{\mathrm{HS}}^2 \leq a^2/\gamma^2, \qquad D_Q \leq a^2/\gamma^2. \tag{19}$$

Explicitly the test has paired value and squared test norm equal to $D_{P,N}$, so $D_{P,N} \le (a/\gamma)\sqrt{D_{P,N}}$; monotone convergence gives the claim. It does not replace the full off-diagonal sequence by its first N entries.

Parseval, including the doubled constant modes, gives

$$I_P := \int \|P + P^*\|_{\mathrm{HS}}^2 \, d\tau = 4D_P, \qquad I_Q = 4D_Q, \tag{20}$$

$$b^2 := \int \|M + L^*\|_{\mathrm{HS}}^2 \, d\tau = 4\|M_0\|_{\mathrm{HS}}^2 + \sum_{k\ge 1} (\|M_k\|_{\mathrm{HS}}^2 + \|L_k\|_{\mathrm{HS}}^2) \ge a^2. \tag{21}$$

For $J = \mathrm{diag}(I_m, -I_m)$, positivity of the compression

$$\Delta = [X \; Y]^*(R + R^*)[X \; Y] = \begin{pmatrix} P + P^* & L + M^* \\ M + L^* & Q + Q^* \end{pmatrix} \succeq 0 \tag{22}$$

implies

$$0 \le \mathrm{tr}(\Delta J \Delta J) = \|P + P^*\|_{\mathrm{HS}}^2 + \|Q + Q^*\|_{\mathrm{HS}}^2 - 2\|M + L^*\|_{\mathrm{HS}}^2.$$

For justification, cyclicity makes the trace $\mathrm{tr}((\Delta^{1/2} J \Delta^{1/2})^2)$, the squared Hilbert–Schmidt norm of a Hermitian matrix. Integration yields $2a^2 \le 2b^2 \le 4(D_P + D_Q) \le 8a^2/\gamma^2$. Division by $a^2 > 0$ finishes the proof. No factor depending on m occurs. □

4 Geometric Closure and the Sharp Endpoint

We record the relevant classical conformal inputs rather than assuming a collar for a nonsmooth numerical range. Riemann mapping, Carathéodory's Jordan-boundary extension, and Schwarz reflection are used in their classical forms [5]; their general proofs are not reproduced.

Lemma 4.1 (Analytic exterior collar)**.** *Every Ω of Section 2 admits h satisfying* (3) *and* (4) *on some univalent collar.*

Proof. Choose $z_* \in \Omega$. Inversion $z \mapsto 1/(z - z_*)$, including infinity, takes the spherical exterior to a bounded simply connected Jordan domain U containing zero, with analytic regular boundary. Convexity implies that each ray from z_* exits once; inversion makes U star-shaped about zero. Let $g : \mathbb{D} \to U$ be a Riemann map with $g(0) = 0$. It extends homeomorphically to the boundary. Analytic regular arcs admit holomorphic coordinates taking them to real intervals. In such charts g has real boundary values and Schwarz reflection continues it. Reflecting g^{-1} as well and applying the identity theorem to the compositions proves nonzero derivative and a local inverse at every boundary point.

For completeness, these local continuations do yield one univalent collar. Take boundary-centered continuation disks of radii R_i whose third-radius disks cover $\mathbb{T}$. If two smaller disks meet, the larger-radius original disk contains the other center: the center distance is less than $(R_i + R_j)/3 \le 2\max(R_i, R_j)/3$. Their connected overlap meets $\mathbb{D}$; the identity theorem glues the continuations. The resulting map is analytic on a neighborhood of $\overline{\mathbb{D}}$. If it failed to be injective on every smaller neighborhood, colliding distinct pairs could be selected with limits in $\overline{\mathbb{D}}$. Boundary injectivity identifies the two limits; a local inverse at their common limit then forbids the collisions. Compactness gives univalence on $|w| < R_0$ for some $R_0 > 1$. Its only zero is the simple zero at zero. Thus $h(\lambda) = z_* + 1/g(1/\lambda)$ has the asserted meromorphic univalent extension with $r = R_0^{-1}$. On $\mathbb{T}$, the radial and angular derivatives are q and iq. Increasing radius enters the exterior, so q is outward. The convex supporting line proves (4). □

Lemma 4.2 (Outer approximation including points and segments)**.** *For every nonempty compact convex $K \subset \mathbb{C}$ there are domains Ω_j as above with $K \subset \Omega_j$ and $\sup_{z\in\overline{\Omega}_j} \operatorname{dist}(z, K) \to 0$.*

Proof. For $0 < \delta < 1$ choose a finite Euclidean δ-net E_δ in $\mathbb{T}$ including $1, -1, i, -i$, and let $s_K(\nu) = \max_K \operatorname{Re}(\bar{\nu}z)$. Choose $t > 0$ with $|E_\delta|e^{-t\delta} < 1$. Define

$$\Phi(z) = \sum_{\nu\in E_\delta} e^{t(\operatorname{Re}(\bar{\nu}z)-s_K(\nu)-\delta)}, \qquad \Omega_\delta = \{\Phi < 1\}.$$

Then $K \subset \Omega_\delta$. The four axis terms enclose $\{\Phi \le 1\}$ in a rectangle independent of $\delta < 1$. Its Hessian is a positive weighted sum of $\nu\nu^{\mathsf{T}}$ and is positive definite because of the axes. A critical point is a global minimum, whereas the minimum is below one; hence the level one is regular and analytic. Its bounded convex interior is nonempty. Radial projection from an interior point maps the boundary homeomorphically onto the circle: on each ray strict convexity and boundedness give exactly one boundary crossing. Thus the boundary is Jordan.

For $z \notin K$ in this sublevel choose a nearest point $w \in K$ and put $d = |z - w|$, $\nu = (z - w)/d$. Choose $\mu \in E_\delta$ with $|\mu - \nu| \le \delta$ and a support maximizer $v \in K$ for μ. Minimality of w implies $\operatorname{Re}(\bar{\nu}(v - w)) \le 0$. Each summand of $\Phi(z) \le 1$ is at most one, so

$$(1 - \delta)d \le \operatorname{Re}(\bar{\mu}(z - w)) \le \operatorname{Re}(\bar{\mu}(v - w)) + \delta \le \delta(1 + \operatorname{diam} K).$$

This proves the uniform distance bound and the result by $\delta \downarrow 0$. □

The compactness of $W(A)$ follows from the unit sphere. Its convexity can be seen without a full-dimensional assumption: compress to the span of two testing vectors. In dimension two a numerical value is a real-affine function of $(2\operatorname{Re}(\bar{u}v), 2\operatorname{Im}(\bar{u}v), |u|^2 - |v|^2)$, which fills the unit sphere in $\mathbb{R}^3$. The linear map from $\mathbb{R}^3$ to $\mathbb{R}^2$ has a kernel direction, so the image of the sphere equals the image of the ball and is convex. Each required segment is therefore in the compressed numerical range and hence in $W(A)$.

Apply Lemmas 4.2 and 4.1 with $K = W(A)$ and scale Theorem 3.1 by $\max_{\overline{\Omega}_j} \|F\|$. If this number is zero, F vanishes identically. All domains lie in one compact rectangle, so uniform continuity of the fixed $\|F(z)\|$ and the distance bound imply convergence of these maxima to $\max_K \|F\|$. This proves the inherited claim (2), including numerical ranges with empty interior. Neither n, m, F nor the coefficient tail cutoff is varying in this limit.

The constant cannot be smaller universally: $A = \left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right)$ has norm two, while $|x^*Ax| = 2|x_1x_2| \le 1$ with equality for equal positive components. For $F(z) = z$ the right-hand maximum is one. This endpoint example will remain compatible with every local refinement below.

5 A Local Defect Identity and Near Extremality

Retain all assumptions and notation of Theorem 3.1, including the *exact top singular pair* and $\gamma > 0$. Define

$$E = b^2 - a^2 = 3\|M_0\|_{\mathrm{HS}}^2 + \sum_{k\ge1} \|L_k\|_{\mathrm{HS}}^2, \tag{23}$$

$$\beta = 4(D_P + D_Q) - 2b^2 = \int_{\mathbb{T}} \|\Delta^{1/2} J \Delta^{1/2}\|_{\mathrm{HS}}^2 \, d\tau, \tag{24}$$

$$\varepsilon_Z = Q_H(Z) - D_Z \quad (Z = P, Q), \tag{25}$$

$$t_Z = a^2/\gamma^2 - Q_H(Z) \quad (Z = P, Q). \tag{26}$$

Here $D_Z = \|Z_0\|_{\mathrm{HS}}^2 + \frac{1}{2}\sum_{k\geq 1}\|Z_k\|_{\mathrm{HS}}^2$ for any sequence, even though the Parseval diagonal interpretation requires a Hermitian Z_0. The letter t_Z in this section denotes an ordered-test defect, not a tail bound. Tail bounds in Section 6 are denoted T_Z.

Theorem 5.1 (Exact source-specific defect accounting)**.** *All quantities $E, \beta, \varepsilon_P, \varepsilon_Q, t_P, t_Q$ are nonnegative. They satisfy the exact identity*

$$2a^2\left(\frac{4}{\gamma^2} - 1\right) = 2E + \beta + 4(\varepsilon_P + \varepsilon_Q) + 4(t_P + t_Q). \tag{27}$$

In particular, with $\mathcal{J} = 2E + \beta + 4(\varepsilon_P + \varepsilon_Q)$,

$$\gamma^2 \leq \frac{4a^2}{a^2 + \mathcal{J}/2} \leq 4. \tag{28}$$

The first upper bound is strictly less than four whenever $\mathcal{J} > 0$. Moreover, writing $u = \overline{Z_+}$,

$$\varepsilon_Z = \frac{1}{2}\langle u, (I - S^*S)(I + S^*S)^{-1}u\rangle. \tag{29}$$

If a certified $\|S\| \leq \sigma < 1$ is available, then

$$\varepsilon_Z \geq \frac{1 - \sigma^2}{2(1 + \sigma^2)}\sum_{k\geq 1}\|Z_k\|_{\mathrm{HS}}^2. \tag{30}$$

Proof. The identities defining E and β were established by Parseval and positive compression. The exact metric version of (18), followed by Proposition 2.2, gives $Q_H(P), Q_H(Q) \leq a^2/\gamma^2$, so $t_P, t_Q \geq 0$. Since $\mathcal{H} \preceq \mathcal{D}$, inversion gives $\mathcal{H}^{-1} \succeq \mathcal{D}^{-1}$, proving $\varepsilon_Z \geq 0$. More explicitly, on the positive modes,

$$(I + S^*S)^{-1} - \tfrac{1}{2}I = \tfrac{1}{2}(I - S^*S)(I + S^*S)^{-1}.$$

Both factors commute because they are functions of the same positive operator, and the constant-mode difference is zero. This proves (29); spectral calculus on $[0, \sigma^2]$ proves (30).

Now use $Q_H(P) + Q_H(Q) = 2a^2/\gamma^2 - (t_P + t_Q)$ and $4(D_P + D_Q) = 2b^2 + \beta = 2a^2 + 2E + \beta$. Substitution gives (27) exactly. Deleting only the nonnegative ordered-test defects and rearranging gives (28). All denominators are strictly positive because $a > 0$. □

Corollary 5.2 (Necessary near-extremality conditions)**.** *For $0 \leq \eta < 1$, suppose $\gamma \geq 2(1 - \eta)$ and put $d_\eta = (1 - \eta)^{-2} - 1$. Then*

$$3\|M_0\|_{\mathrm{HS}}^2 + \sum_{k\geq 1}\|L_k\|_{\mathrm{HS}}^2 \leq a^2 d_\eta, \tag{31}$$

$$\beta \leq 2a^2 d_\eta, \tag{32}$$

$$\varepsilon_P + \varepsilon_Q \leq \tfrac{1}{2}a^2 d_\eta, \qquad t_P + t_Q \leq \tfrac{1}{2}a^2 d_\eta. \tag{33}$$

Under $\|S\| \leq \sigma < 1$, additionally

$$\sum_{k\geq 1}(\|P_k\|_{\mathrm{HS}}^2 + \|Q_k\|_{\mathrm{HS}}^2) \leq a^2 d_\eta \frac{1 + \sigma^2}{1 - \sigma^2}. \tag{34}$$

Proof. The left side of (27) is at most $2a^2 d_\eta$. Each displayed bound follows by dropping the other nonnegative terms. For (34) combine (33) with (30). No compactness argument or unproved equality classification is needed. □

At exact equality $\gamma = 2$, these formulas force $M_0 = 0$, $L_k = 0$ for $k \geq 1$, and $\Delta^{1/2} J \Delta^{1/2} = 0$ almost everywhere, together with zero Gram and ordered-test defects. They are necessary conditions, not a classification or a claim of attainability for a fixed strictly enclosing domain. In a shrinking-domain family, the collar, S, a, and the singular pair can all change. Passing these quantitative statements to that limit requires additional uniform estimates; only the universal inequality was passed to the limit in Section 4.

5.1 What this refinement does and does not compute

The bound uses singular vectors of the evaluated matrix. It is a diagnostic and certification identity, not by itself a cheaper algorithm to discover $\|F[A]\|$. Arbitrary test vectors, or a merely approximately stationary pair, do not satisfy (17). A separate perturbation analysis would be required for approximate pairs. The argument is uniform in coefficient dimension because S acts on scalar mode indices and each matrix entry identically; this does not assert dimension-independent computational cost.

6 Finite Sections Without Discarding Tails

The following results are conditional certificates: the constants and finite data must themselves be rigorously enclosed. Sampled boundary maxima and floating-point singular values without error bounds are not such enclosures.

Proposition 6.1 (Two distinct tail bounds)**.** *Suppose the known bounds*

$$|s_{kj}| \leq C_s \rho^{k+j} \quad (0 < \rho < 1), \qquad \|b_k(A)\| \leq C_R \theta^k \quad (0 < \theta < 1) \tag{35}$$

hold for all indicated indices. Let Π_N retain modes $1, \ldots, N$, with $N \geq 1$, and $S_N = \Pi_N S \Pi_N$ extended by zero. Then

$$\|S - S_N\| \leq e_N := \frac{C_s \rho^2}{1 - \rho^2} \sqrt{2\rho^{2N} - \rho^{4N}}, \tag{36}$$

$$\left(\sum_{k>N} \|Z_k\|_{\mathrm{HS}}^2 \right)^{1/2} \leq T := \frac{C_R \theta^{N+1}}{\sqrt{1 - \theta^2}} \quad (Z = P, Q, M, L). \tag{37}$$

For $\mathcal{H}_N = 1 \oplus (I + S_N^* S_N)$ on the full sequence space,

$$\|\mathcal{H}^{-1} - \mathcal{H}_N^{-1}\| \leq 2 e_N. \tag{38}$$

Proof. The square of the Hilbert–Schmidt norm of $S - S_N$ is the sum of $|s_{kj}|^2$ outside the N by N square. With $r = \rho^2$, it is bounded by

$$C_s^2 \left\{ \left(\frac{r}{1 - r} \right)^2 - \left(\frac{r(1 - r^N)}{1 - r} \right)^2 \right\} = \frac{C_s^2 \rho^4}{(1 - \rho^2)^2} (2\rho^{2N} - \rho^{4N}).$$

Operator norm is at most Hilbert–Schmidt norm. For every block, for example $M_k = Y^* b_k(A) X$, one has $\|M_k\|_{\mathrm{HS}} \leq \|Y\| \, \|b_k(A)\| \, \|X\|_{\mathrm{HS}} \leq C_R \theta^k$. The same argument treats the others; summing the squared geometric tail proves (37). Since $\|S_N\| \leq \|S\| \leq 1$, $\|S^* S - S_N^* S_N\| \leq 2 e_N$. Both Gram operators dominate I, so their inverse norms are at most one. The resolvent identity for the inverses then proves (38). □

For a sequence Z, let $Z^{[N]}$ retain modes $0,\dots,N$, and suppose a bound $\|Z - Z^{[N]}\| \le T_Z$ is given (the common T above is one choice). Define finite computable quantities

$$q_{Z,N} = \langle \overline{Z^{[N]}}, \mathcal{H}_N^{-1} \overline{Z^{[N]}} \rangle, \tag{39}$$

$$\delta_{Z,N} = 2e_N \|Z^{[N]}\|^2 + 2\|Z^{[N]}\| T_Z + T_Z^2, \tag{40}$$

$$D_{Z,N} = \|Z_0\|_{\mathrm{HS}}^2 + \tfrac{1}{2} \sum_{k=1}^{N} \|Z_k\|_{\mathrm{HS}}^2. \tag{41}$$

The inverse in (39) only requires a finite solve, since the omitted part of $\mathcal{H}_N$ is identity.

Theorem 6.2 (A certified finite defect bound)**.** *Under the exact analytic-domain and singular-pair hypotheses of Theorem 5.1, the tail hypotheses of Proposition 6.1, and valid (possibly sharper) bounds T_Z, set*

$$A_N = \sum_{k=0}^{N} \|M_k\|_{\mathrm{HS}}^2, \qquad U = A_N + T_M^2, \tag{42}$$

$$B_N = 4\|M_0\|_{\mathrm{HS}}^2 + \sum_{k=1}^{N} (\|M_k\|_{\mathrm{HS}}^2 + \|L_k\|_{\mathrm{HS}}^2), \tag{43}$$

$$E_N = 3\|M_0\|_{\mathrm{HS}}^2 + \sum_{k=1}^{N} \|L_k\|_{\mathrm{HS}}^2, \tag{44}$$

$$\beta_- = \max\{0,\ 4(D_{P,N} + D_{Q,N}) - 2B_N - 2(T_M^2 + T_L^2)\}, \tag{45}$$

$$\varepsilon_{Z,-} = \max\{0,\ q_{Z,N} - \delta_{Z,N} - D_{Z,N} - T_Z^2/2\}, \quad Z = P, Q, \tag{46}$$

$$\mathcal{J}_- = 2E_N + \beta_- + 4(\varepsilon_{P,-} + \varepsilon_{Q,-}). \tag{47}$$

Then $U > 0$ and

$$\gamma^2 \le \frac{4U}{U + \mathcal{J}_-/2} \le 4. \tag{48}$$

For exact finite coefficients and tail bounds tending to zero, these finite lower defects converge to $\mathcal{J}$, and $U \to a^2$.

Proof. For any contraction $V \succeq 0$ and vectors $z = z_N + z_T$, $|\langle z, Vz\rangle - \langle z_N, Vz_N\rangle| \le 2\|z_N\|\|z_T\| + \|z_T\|^2$. Apply this with $V = \mathcal{H}^{-1}$ and then (38) to obtain $|Q_H(Z) - q_{Z,N}| \le \delta_{Z,N}$. Also $D_Z \le D_{Z,N} + T_Z^2/2$; this proves $0 \le \varepsilon_{Z,-} \le \varepsilon_Z$.

The full cross energy is at most $B_N + T_M^2 + T_L^2$, whereas $D_P + D_Q \ge D_{P,N} + D_{Q,N}$. Thus $0 \le \beta_- \le \beta$. Direct positive sums give $E_N \le E$ and $0 < a^2 \le U$, so $0 \le \mathcal{J}_- \le \mathcal{J}$. The function $4u/(u + j/2)$ is increasing in $u > 0$ and decreasing in $j \ge 0$. Apply these two monotonicities to (28). Finally all four sequence tails vanish, $e_N \to 0$, and their retained norms stay bounded. Hence $\delta_{Z,N} \to 0$. Each unclamped expression converges to its nonnegative exact defect; continuity of $\max(0,\cdot)$ proves the last assertion. □

The two tail mechanisms are independent. Small $S - S_N$ says nothing by itself about the compressed resolvent coefficients beyond N. Conversely, small M-tail does not justify replacing the Faber metric by an identity matrix. Neither cut discards coefficients of FG or GF in the inherited ordered-test proof.

6.1 The exact retained/complement inverse

There is another useful way to expose what a finite Gram inverse omits. For the positive-mode Gram $H_+ = I + S^*S$, split the coefficient space into the first N modes and their complement, and write

$$H_+ = \begin{pmatrix} A_0 & B_0 \\ B_0^* & C_0 \end{pmatrix}, \qquad K_0 = C_0 - B_0^* A_0^{-1} B_0.$$

The symbols in this display are Gram blocks, not the base matrix A or polynomial coefficients.

Proposition 6.3 (Exact Schur complement contribution)**.** *One has $A_0 \succeq I$, $K_0 \succeq I$. For $u = \bar{Z}_+ = u_N \oplus u_T$,*

$$Q_H(Z) = \|Z_0\|_{\mathrm{HS}}^2 + \langle u_N, A_0^{-1} u_N \rangle + \langle u_T - B_0^* A_0^{-1} u_N, K_0^{-1}(u_T - B_0^* A_0^{-1} u_N) \rangle. \tag{49}$$

In particular the first two terms give a valid lower bound, but generally not equality even when $u_T = 0$.

Proof. The finite principal block inherits $A_0 \succeq I$. For each tail vector v, minimize $\langle (w, v), H_+(w, v) \rangle$ over w. The minimizing vector is $w = -A_0^{-1} B_0 v$, and the minimum is $\langle v, K_0 v \rangle$. Since $H_+ \succeq I$, that minimum is at least $\|v\|^2$. Thus K_0 is bounded, positive, and invertible on the tail Hilbert space. The factorization

$$H_+ = \begin{pmatrix} I & 0 \\ B_0^* A_0^{-1} & I \end{pmatrix} \begin{pmatrix} A_0 & 0 \\ 0 & K_0 \end{pmatrix} \begin{pmatrix} I & A_0^{-1} B_0 \\ 0 & I \end{pmatrix}$$

is verified by block multiplication. Inverting and taking its quadratic form proves (49). □

Crucially $A_0 = I + \Pi_N S^* S \Pi_N$ contains *all output rows* of S. It is not generally $I + S_N^* S_N$. Omitting those rows makes the finite Gram smaller and its inverse larger; such an inverse cannot silently replace A_0^{-1} in a certified lower bound. Proposition 6.1 supplies one valid error allowance. Exact Schur algebra is classical; its contribution here is specifying which infinite Faber terms and bilinear conjugations the certificate actually requires.

7 Exact Examples and Failure Tests

7.1 Disk: a sharp limiting example with retained slack

Take $A = \left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right)$, $\Omega = \{|z| < R\}$ for $R > 1$, $h(\lambda) = R\lambda$, and $F(z) = z/R$. Then $b_k(z) = (z/R)^k$, $S = 0$, $\mathcal{H} = I$ and $\gamma = 2/R$. Choose $X = e_2$, $Y = e_1$ at scalar coefficient level. Direct multiplication gives

$$P_0 = Q_0 = 1, \quad M_1 = 2/R, \quad M_0 = L_0 = 0, \quad P_k = Q_k = L_k = 0 \ (k \geq 1), \quad M_k = 0 \ (k \geq 2).$$

Consequently

$$a^2 = b^2 = 4/R^2, \quad E = 0, \quad \beta = 8 - 8/R^2, \quad \varepsilon_P = \varepsilon_Q = t_P = t_Q = 0.$$

Equation (28) is exact: $\gamma^2 \leq 4/R^2$. The positive block slack is genuinely useful for a fixed enclosing disk, but tends to zero as $R \downarrow 1$. Thus the refinement respects the universal sharp endpoint instead of purporting to improve it.

7.2 Ellipse: a nontrivial exact Gram with infinite tails

For $0 < \alpha < 1$, let $h(\lambda) = \lambda + \alpha/\lambda$. If $h(\lambda) = h(\mu)$, then either $\lambda = \mu$ or $\lambda\mu = \alpha$. It follows that h is univalent on $|\lambda| > \sqrt{\alpha}$; the circle maps to the ellipse with semiaxes $1 + \alpha$ and $1 - \alpha$. Direct calculation gives

$$s(\lambda, \mu) = \frac{\alpha}{\lambda\mu - \alpha}, \quad s_{kj} = \alpha^k \mathbf{1}_{\{k=j\}}, \quad \mathcal{H} = \operatorname{diag}(1, 1 + \alpha^2, 1 + \alpha^4, \ldots). \tag{50}$$

Also

$$b_1(z) = z, \quad b_2(z) = z^2 - 2\alpha, \quad b_k(z) = z b_{k-1}(z) - \alpha b_{k-2}(z) \quad (k \geq 3),$$

as follows by multiplying (5) by $1 - z/\lambda + \alpha/\lambda^2$; the special $k = 2$ term must not be replaced by the ordinary recurrence with $b_0 = 1$. Here the Gram defect is exactly

$$\varepsilon_Z = \sum_{k \geq 1} \frac{1 - \alpha^{2k}}{2(1 + \alpha^{2k})} \|Z_k\|_{\mathrm{HS}}^2.$$

For example $A = 0$ at scalar base and coefficient levels, with $X = Y = 1$, has $P_{2j} = 2(-\alpha)^j$, $P_{2j+1} = 0$ for $j \geq 1$, and $P_0 = 1$. There are infinitely many nonzero coefficients even though A is scalar and nilpotent. For a cutoff $N = 2J$ the squared omitted norm is $4\alpha^{2(J+1)}/(1 - \alpha^2)$, not zero. The same example has strictly positive Gram defect. It disproves any inference that finite base dimension, nilpotence, or polynomial test degree makes the resolvent Faber stream finite.

7.3 Why the two ordered tests cannot be merged

The coefficient matrices need not commute. For $D = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $E = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, $DE = \operatorname{diag}(1, 0)$ whereas $ED = \operatorname{diag}(0, 1)$. For $F(z) = zD$, $G(z) = E$, their evaluated products are $A \otimes DE$ and $A \otimes ED$. Only the singular-pair identities (17) control the two different diagonal arrays. A scalar commutativity lemma cannot simply be invoked after replacing scalar coefficients by these matrices.

8 Private Proof Lineage and Source-Specific Transfer

The catalogue's L4 screening was an applicability judgment, not a proof or contribution score. The following actual original proofs were read. They do not assert the complete Crouzeix theorem.

In TR-core-pat-4 [6], the restricted dual theorem uses a physical Gram G and observation metric W, with $A_h = W^{1/2} T_h G^{-1/2}$. Full column rank gives $D_h T_h = I$, and an unresolved orthogonal component contributes its own squared norm to the reconstruction error. The proof of its finite weak-test formula $b^* H^{-1} b$ makes that quantity a *lower*, not upper, bound for a full dual norm unless the complement is controlled. The coercive PDE residual theorem uses its own ellipticity and boundary assumptions and does not transfer to the nonnormal matrix functional calculus.

In TR-core-pat-5 [7], the Neumann proof bounds each power by a specified induced norm, not merely the spectral radius. Its reconstruction theorem keeps the retained spectral space explicit. The Schur residual proof retains eliminated source and residual terms. The Woodbury proof assumes invertibility on the same fixed coordinate space. These are standard algebraic mechanisms with precise contracts, not substitutes for the positive exterior kernel.

The physical-metrics chapter [8] proves $G = J^* J$, $K = J^* A J$, $B = G^{-1} K$ and the exact residual decomposition

$$\|(A - \lambda) J c\|^2 = \|(B - \lambda) c\|_G^2 + \|(I - J G^{-1} J^*) A J c\|^2.$$

It also gives a concrete failure of entrywise-small Gram perturbations to control relative metric error. We use its distinction between an actual norm and raw coordinates, not its self-adjoint compression hypotheses.

The specific transfer here is $\mathcal{H} = 1 \oplus (I + S^*S)$ as the actual boundary norm, the conjugated array $\bar{Z}$ as its dual data, and the two separate tails in Proposition 6.1. Positivity of K and the two ordered products remain indispensable new problem-specific inputs from the public source. The defect identity and finite bound are proved above rather than attributed to the private reports. No other private corpus, patent history, or transitive references are claimed to have been read.

9 Assessment, Limits, and Next Mathematical Steps

The strongest feature of the source is that the amplification is explicit: arbitrary m, unnormalized Hilbert–Schmidt coefficients, and both ordered products are present before the final positive-block comparison. Convexity is used geometrically, and neither symmetry of s_{kj} nor equality of row and column operator norms is assumed. The source also avoids an unjustified limiting collar for a segment or point.

The new quantitative result retains three losses in precisely that argument. The near-extremality corollary is stronger information than the universal bound, and the finite certificate states how much uncomputed coefficient energy can be tolerated. This supports C3 as a scoped analytic synthesis, not C4 or C5: the universal theorem and its main mechanism are inherited, the underlying norm and Schur identities are conventional, and no computational speedup, novelty priority, or independently accepted solution of the general conjecture is established.

Practical limitations are substantive. An analytic exterior map and certified collar bounds may be costly to obtain. ρ, θ may approach one as the enclosure shrinks, making the displayed tails expensive. Exact singular-pair data are assumed, so the present certificate is not immediately an interval implementation for approximate singular vectors. A coarse tail allowance may clamp every lower defect to zero, correctly returning only the inherited constant two. This is failure to sharpen, not evidence of a counterexample.

Useful further work would quantify singular-pair residuals in the two ordered identities, construct interval enclosures for conformal coefficients on explicit domains, and determine when near-extremal defect estimates imply geometric or algebraic rigidity. Such claims would require new proofs. Likewise extending the present finite-dimensional polynomial statement to unbounded operators or varying amplification limits is outside this report.

A Literal Read Scope and Provenance

The source directory is `preprints/A-direct-proof-of-the-complete-Crouzeix-inequality-September-26-2026/build`. The actual local files, not just abstracts or catalogue summaries, were read:

File or original	Actual read scope
Public `main.tex`	All lines 1–623, including introduction, Faber proof 182–278, resolvent 295–321, amplified proof 323–453, outer approximation 457–511, limiting proof 513–530, collar 536–621.
Public auxiliaries	Entire `references.tex`, entire `figures/normal.tex` (schematic source), entire README. The public source PDF was not separately page-audited.
TR-core-pat-4	Lines 1–50 (metadata), 139–442, 638–699, 1851–1888. These include the full relevant dual, panel, coercive residual, and weak-test proofs and their hypotheses. Other sections were not fully read.
TR-core-pat-5	Lines 1–38 (metadata), 70–133, 189–466, 855–862. Full relevant Neumann, frame, Schur residual and Woodbury proofs, with surrounding assumptions. Other sections were not fully read.
Volume 21 chapter	All 210 lines of `ch01d-physical-metrics.tex`, including its exact examples and limitations.

The catalogue M630 row and the frozen upstream M630 identity records were read as metadata. Public arXiv version pages and relevant HTML theorem and scope passages were inspected; this is not a full proof audit of those external papers. In particular the Lorist–Schwenninger main argument was read for the scalar/complete distinction, whereas the much longer Åhag–Czyż–Virtanen paper was not read in its entirety.

The following SHA-256 values bind the actual source bytes. The report's own final hash is recorded externally in the contribution assessment and audit manifest, avoiding a self-referential hash claim.

Public main: `54a429fe7129b4760b3015fabf2269d4de`
`5fa38aba9539b18542c1df7e58333f`

Public references: `6402be83926f774063d5e4e65a0075a686`
`82419f388eb9c7c500231fb350b2b6`

Public figure source: `3e6845a1434a17fa3630036fa2b38337fc`
`232253afcdb43cf3ee7c185b558391`

Private TR-core-pat-4: `6e2b2b3c29914c42134ddfa9d884615c`
`3c373cae2ffea846d6326c5dd9ac996f`

Private TR-core-pat-5: `8e7250be4aa3874b2d03386a78cbe6ff`
`c432785469e3af70924e5d93d9bf5183`

Private physical-metrics chapter: `7bc5898243373ee2e45551aa78991808`
`fa020148dc6d07c91c41351b5bb84c5c`

The public main Git blob is `5976502d77febf477933f36f6205c6e1aa6370ae`. Source identity is not proof validation. Private content was not submitted to any public search or external service; external requests used public titles, arXiv identifiers, and public repository URLs only.

B Verification Contract

The companion local audit stores the executed verification and export results bound to final file hashes. The test script checks exact rational disk identities, the ellipse recurrence and geometric tail, noncommuting tensor products and bilinear conjugation, and exact finite Schur algebra. Floating-point supplementary tests exercise complex Faber metrics and finite-cutoff bounds; they are regression checks, not interval certificates or an independent proof of infinitely many cases. The

analytic proofs above do not rely on those samples. The exported PDF is visually inspected in full, separately from the mathematical checks. The executed run passed 48 exact algebra checks, 120 floating-point regression checks, and 12 document/provenance checks. The native editor/compiler tools were not exposed in this session; the authorized existing local export helper used the installed MiKTeX and Poppler, without installing software. This tool-availability limitation is not represented as a successful native compile.

References

[1] OpenAI, *A direct proof of the complete Crouzeix inequality*, September 26, 2026, OpenAI Math Release preprint. Pinned public source. Actual read-file hashes appear in Appendix A.

[2] E. Lorist and F. L. Schwenninger, *A solution to Crouzeix's conjecture*, arXiv:2608.03841v2, August 17, 2026. `https://arxiv.org/abs/2608.03841v2`.

[3] P. Åhag, R. Czyż, and J. Virtanen, *The Complete Crouzeix Conjecture in Dimension Three and the Clouâtre–Ostermann–Ransford conjecture*, arXiv:2608.27346v4, September 21, 2026. `https://arxiv.org/abs/2608.27346v4`.

[4] M. Crouzeix and C. Palencia, *The numerical range is a* $(1+\sqrt{2})$*-spectral set*, SIAM J. Matrix Anal. Appl. **38** (2017), 649–655. `https://doi.org/10.1137/17M1116672`.

[5] L. V. Ahlfors, *Complex Analysis*, 3rd ed., McGraw–Hill, 1979. Classical conformal mapping, boundary correspondence, and reflection inputs; not an assertion that the whole book was reread.

[6] M. Subbarao, *Measure-Aware Neural PDE Operators with Reconstructible Tokens and Residual Gates*, TR-core-pat-4, October 1, 2026. Confidential local original; actual selected proof ranges in Appendix A.

[7] M. Subbarao, *Adaptive Local–Global Graph Operators with Certified Updates and Reconstruction*, TR-core-pat-5, October 1, 2026. Confidential local original; actual selected proof ranges in Appendix A.

[8] M. Subbarao, *A Nonorthogonal Representation Carries a Metric*, Volume 21 local chapter `ch01d-physical-metrics.tex`. Confidential local original, all 210 lines read; no external publication status inferred.